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LIMITING BEHAVIOUR OF THE APPROXIMATE FIRST ORDER AND SECOND ORDER DIRECTIONAL DERIVATIVES FOR A CONVEX FUNCTION

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INTRODUCTION

THE APPROXIMATE first order directional derivative of a convex function has been proved to be a useful tool in convex analysis, from the theoretical viewpoint as well as from the computational viewpoint. Given a convex function f defined on $\mathbb{R}^n$, the approximate first order directional derivative of f at x_0 in the direction d, denoted as $f'_\varepsilon(x_0; d)$, enjoys some noteworthy properties different as for their nature from those of the 'exact' directional derivative $f'(x_0; d)$. The function $v_{d,\varepsilon}: x \mapsto f'_\varepsilon(x; d)$ turns out to be locally Lipschitz whenever $\varepsilon > 0$ [15, 10] and, at those points x_0 where it is not differentiable, $v_{d,\varepsilon}$ admits however a directional derivative for all δ, which we shall denote by $f''_\varepsilon(x_0; d, \delta)$. The study of $v_{d,\varepsilon}$ as a nonsmooth function of x has been carried out in the author's earlier paper [13, sections II and III]; the reader will find there a classification in every detail of all possible situations with regard to the differentiability properties of $v_{d,\varepsilon}$. Unlike that, the present work is devoted to the study of the behaviours of $f'_\varepsilon(x_0; d)$ and $f''_\varepsilon(x_0; d, d)$ as functions of ε, with a particular emphasis on the limiting behaviours when $\varepsilon \to 0^+$. In that respect, earlier results by the author [13, section IV] are improved and completed.

The behaviour of $\theta_d: \varepsilon \mapsto f'_\varepsilon(x_0; d)$ in a neighbourhood of 0^+ is of importance if one wishes to get a better insight into the rate of approximation of $f'(x_0; d)$ by $f'_\varepsilon(x_0; d)$. It is known that θ_d is an increasing function and even that it is locally Lipschitz on $\mathbb{R}^*_+$ [10]. We shall go further in the study of θ_d in section 2 of the present paper. In particular, we shall see that θ_d is a concave function whose superdifferential at all $\varepsilon \geqslant 0$ as well as the conjugate function $(-\theta_d)^*$ can be fully described. The question of the limiting behaviour of $f'_\varepsilon(x_0; d)$ when $\varepsilon \to 0^+$ can partly be answered when knowing the structure of the superdifferential of θ_d at 0. However, we shall go further by tackling questions like: does there exist $\sigma > 0$ such that $f'_\varepsilon(x_0; d) - f'(x_0; d) = o(\varepsilon_\sigma)$?

As aforesaid, the approximate second order directional derivative

$$f''_\varepsilon(x_0; d, d) = \lim_{\alpha \to 0^+} \frac{f'_\varepsilon(x_0 + \alpha d; d) - f'_\varepsilon(x_0; d)}{\alpha}$$

exists whenever $\varepsilon > 0$, and the exact expression of $f''_\varepsilon(x_0; d, d)$ [14, 1] involves only the behaviour of f on the half-line $x_0 + \mathbb{R}_+ d$. A nice feature of $f''_\varepsilon(x_0; d, d)$ is that it is defined at

all x_0 and for all d. However, a fundamental question which has to be answered is the following: to what extent is $f''_\varepsilon(x_0; d, d)$ an approximation of the second order directional derivative of f at x_0 in the d direction? This question was firstly approached by the author in [13, section IV] where, in particular, it was proved that $\lim_{\varepsilon \to 0^+} f''_\varepsilon(x_0; d, d)$ exists uniformly in d for almost all x_0. By imposing a mild assumption on the behaviour of f on the half-line $x_0 + \mathbb{R}_+ d$, we shall prove in section III that $\lim_{\varepsilon \to 0^+} f''_\varepsilon(x_0; d, d)$ does exist and equals the second order directional derivative of f at x_0 in the d direction. It is noteworthy that such a limit exists at all x_0 and for all d when f is of the (discrete) minimax type, i.e.,

$$f(x) = \max\{f_i(x) \mid i = 1, \ldots, m\},$$

where the f_i are convex twice continuously differentiable functions. In addition to that, results on the (generalized) derivative of the function $\rho_d : \varepsilon \mapsto f''_\varepsilon(x_0; d, d)$ at $\varepsilon > 0$ will be provided.

1. DEFINITIONS AND PRELIMINARY PROPERTIES

Throughout the sequel, f will be a convex function from $\mathbb{R}^n$ into $\mathbb{R}$. Given such an f, the ε-subdifferential of f at $x_0 \in \mathbb{R}^n$ is defined for each $\varepsilon \geqslant 0$ as the set of vectors $x^* \in \mathbb{R}^n$ satisfying

$$f(x) \geqslant f(x_0) + \langle x^*, x - x_0 \rangle - \varepsilon \qquad \text{for all } x \in \mathbb{R}^n. \tag{1.1}$$

The set of such vectors, denoted by $\partial_\varepsilon f(x_0)$, is a nonempty compact convex set which reduces to the subdifferential $\partial f(x_0)$ when $\varepsilon = 0$. To suppose that f is finite everywhere is just a technical assumption which makes the notations simpler. All the results we shall display throughout hold true for a lower-semicontinuous convex function from $\mathbb{R}^n$ into $(-\infty, +\infty]$, provided that the considered x_0 lies in the interior of domf.

Let us recall the characterization of $\partial_\varepsilon f(x_0)$ by means of its support function.

PROPOSITION 1.1. [16, p. 220]. The support function of $\partial_\varepsilon f(x_0)$ is given as

$$d \mapsto f'_\varepsilon(x_0; d) = \inf_{\lambda > 0} \{[f(x_0 + \lambda d) - f(x_0) + \varepsilon]\lambda^{-1}\}. \tag{1.2}$$

From now on, x_0 is fixed and d is a non-null direction. For all $\varepsilon \geqslant 0$, we denote by $\Lambda_d(\varepsilon)$ the (possibly empty set) of $\lambda_0 > 0$ for which

$$[f(x_0 + \lambda_0 d) - f(x_0) + \varepsilon]\lambda_0^{-1} = f'_\varepsilon(x_0; d).$$

The behaviour of the (approximate) difference quotient

$$q_{d,\varepsilon} : \lambda \mapsto [f(x_0 + \lambda d) - f(x_0) + \varepsilon]\lambda^{-1}$$

on $\mathbb{R}^*_+$ is of particular importance for deriving the properties of $\Lambda_d(\varepsilon)$. When $\varepsilon = 0$, we simply denote $q_{d,\varepsilon}$ by q_d. The behaviour of the $q_{d,\varepsilon}$ near 0^+ and $+\infty$ is known since

$$\lim_{\lambda \to 0^+} q_d(\lambda) = \inf_{\lambda > 0} q_d(\lambda) = f'(x_0; d),$$

$$\lim_{\lambda \to 0^+} q_{d,\varepsilon}(\lambda) = +\infty \quad \text{for all } \varepsilon > 0,$$

and

$$\lim_{\lambda \to +\infty} q_{d,\varepsilon}(\lambda) = \sup_{\lambda > 0} \{[f(x_0 + \lambda d) - f(x_0)]\lambda^{-1}\} = f_\infty(d) \qquad \text{for all } \varepsilon \geqslant 0.$$

Here f_∞ is what is known as the recession function of f (or the asymptotic function of f).

The case $\varepsilon = 0$ deserves some more attention. q_d is known to be an increasing function of $\lambda > 0$ and $\Lambda_d(0)$ can be described as $]0, a_d^*]]$† where a_d^* is the supremum of all $a \in \mathbb{R}_+$ for which

$$f(x_0 + \lambda d) = f(x_0) + \lambda f'(x_0; d) \qquad \text{for all } \lambda \in [0, a].$$

$a_d^* = +\infty$, i.e., $\Lambda_d(0) = \mathbb{R}_+^*$, corresponds precisely to the situation where $f_d : \lambda \mapsto f(x_0 + \lambda d)$ is an affine function on $\mathbb{R}_+$. Having $a_d^* = 0$, i.e., $\Lambda_d(0) = \varnothing$, amounts to saying that

$$f(x_0 + \lambda d) > f(x_0) + \lambda f'(x_0; d) \qquad \text{for all } \lambda > 0.$$

As for $\varepsilon > 0$, we clearly have that $\Lambda_d(\varepsilon) \subset \mathbb{R}_+^*$ and $\Lambda_d(\varepsilon)$ is nonempty if and only if

$$\exists \lambda_* > 0, [f(x_0 + \lambda_* d) - f(x_0) + \varepsilon]\lambda_*^{-1} \leqslant f_\infty(d). \tag{1.3}$$

It turns out to be useful to look at the $q_{d,\varepsilon}$ as functions of λ^{-1}. More precisely, let $r_{d,\varepsilon}$ be defined for $\varepsilon \geqslant 0$ by

$$r_{d,\varepsilon}(\mu) = q_{d,\varepsilon}\left(\frac{1}{\mu}\right) \qquad \text{for all } \mu > 0. \tag{1.4}$$

Here again, we set $r_{d,\varepsilon} = r_d$ when $\varepsilon = 0$. Rather surprisingly, $r_{d,\varepsilon}$ is a *convex* function on $\mathbb{R}_+^*$. This property was noticed by Lemaréchal and Nurminskii [14] (cf. also Auslender [1]) as a by-product of a duality result. It also can be viewed as a mere consequence of the following: $r(\mu)$ is convex on $\mathbb{R}_+^*$ if and only if $\mu r(1/\mu)$ is convex on $\mathbb{R}_+^*$ (see for example Bourbaki [2, p. 54]). Since $r_{d,\varepsilon}(\mu)$, $\varepsilon \geqslant 0$, converges to $f_\infty(d)$ when $\mu \to 0^+$, we agree on posing

$$r_{d,\varepsilon}(0) = f_\infty(d) \qquad \text{whenever } f_\infty(d) < +\infty.$$

So, r_d is a decreasing convex function of μ with

$$\lim_{\mu \to +\infty} r_d(\mu) = f'(x_0; d).$$

As for the $r_{d,\varepsilon}$, $\varepsilon > 0$, we already have noticed that

$$\lim_{\mu \to +\infty} r_{d,\varepsilon}(\mu) = +\infty,$$

while their characteristics from the viewpoint of convex analysis are easily derived from those of r_d. At this stage, it is quite natural to ask the following questions: what is the conjugate function of r_d? what is the subdifferential of r_d? These questions will be answered in the next section, in connection with the study of $f_\varepsilon'(x_0; d)$ as a function of ε.

Unlike $q_{d,\varepsilon}$, $r_{d,\varepsilon}$ always achieves its minimum value $(= f_\varepsilon'(x_0; d))$ on $\mathbb{R}_+$ whenever $\varepsilon > 0$. If we denote by $M_d(\varepsilon)$ the set of $\mu_0 \in \mathbb{R}_+$ for which

$$r_{d,\varepsilon}(\mu_0) = f_\varepsilon'(x_0, d),$$

$M_d(\varepsilon)$ is, for all $\varepsilon > 0$, a nonempty compact interval of $\mathbb{R}_+$. As to $M_d(0)$, it can be described

† When $\alpha, \beta \in \overline{\mathbb{R}}$, it is convenient to pose

$$]\alpha, \beta]] = \{\lambda \in \mathbb{R} \mid \alpha < \lambda \leqslant \beta\}.$$

A similar convention holds for $[\![\alpha, \beta[$, etc.

as the segment

$$[1/a_d^*, +\infty[\ (1/+\infty = 0 \text{ and } 1/0 = +\infty \text{ by convention}). \tag{1.5}$$

Due to the relationship between $q_{d,\varepsilon}$ and $r_{d,\varepsilon}$, $\mu_0 \in \mathbb{R}_+^*$ is a minimum of $r_{d,\varepsilon}$ on $\mathbb{R}_+^*$ if and only if $\lambda_0 = 1/\mu_0$ is a minimum of $q_{d,\varepsilon}$ on $\mathbb{R}_+^*$. Hence

$$\Lambda_d(\varepsilon) = \left\{\frac{1}{\mu_0} \,|\, \mu_0 \in M_d(\varepsilon), \mu_0 > 0\right\} \qquad \text{for all } \varepsilon \geqslant 0. \tag{1.6}$$

The non-null elements of $M_d(\varepsilon)$ can be characterized in the following way.

PROPOSITION 1.2. $\mu_0 \neq 0$ belongs to $M_d(\varepsilon)$ if and only if

$$0 \in -\frac{1}{\mu_0} < \partial f\left(x_0 + \frac{d}{\mu_0}\right), d > + f\left(x_0 + \frac{d}{\mu_0}\right) - f(x_0) + \varepsilon. \tag{1.7}$$

Proof. A necessary and sufficient condition for $\mu_0 \neq 0$ to be a minimum of $r_{d,\varepsilon}$ on $\mathbb{R}_+$ is that $0 \in \partial r_{d,\varepsilon}(\mu_0)$, where $\partial r_{d,\varepsilon}(\mu_0)$ stands for the subdifferential of $r_{d,\varepsilon}$ at μ_0. Now, since $r_{d,\varepsilon}$ is the product of two locally Lipschitz functions around μ_0, chain rules on generalized gradients in Clarke's sense [4, 6, 8] yield that

$$\partial r_{d,\varepsilon}(\mu_0) = -\frac{1}{\mu_0} < \partial f\left(x_0 + \frac{d}{\mu_0}\right), d > + f\left(x_0 + \frac{d}{\mu_0}\right) - f(x_0) + \varepsilon.$$

Hence the result is proved. ■

The different properties of $\Lambda_d(\varepsilon)$ [or, equivalently, those of $M_d(\varepsilon)$] depend on the range of the function $q_{d,\varepsilon}$ as well as on the value taken by $f_\infty(d)$. The classification of all the possible situations was carried out in [13, section II], in conjunction with the differentiability properties of the function $v_{d,\varepsilon}: x \mapsto f_\varepsilon'(x; d)$. We recall here the fundamental situations to be taken into account with regard to the behaviour of $f_\varepsilon'(x_0; d)$ and $\Lambda_d(\varepsilon)$ when $\varepsilon \to 0^+$. In the first place, we examine the somewhat degenerate case where $a_d^* = +\infty$, i.e. f_d is affine on $\mathbb{R}_+$. This situation can be described by one of the following equivalent statements:

(i) $f_\varepsilon'(x_0; d) = f'(x_0; d)$ for all $\varepsilon > 0$;
(ii) $\{q_{d,\varepsilon}(\lambda) | \lambda \in \mathbb{R}_+^*\} \cap]-\infty, f_\infty(d)] = \varnothing$ for all $\varepsilon > 0$;
(iii) $\Lambda_d(\varepsilon) = \varnothing$ for all $\varepsilon > 0$;
(iv) $M_d(\varepsilon) = \{0\}$ for all $\varepsilon > 0$.

Consider now the situations where $a_d^* < +\infty$. In such a case, there necessarily exists $\bar{\varepsilon} > 0$ for which

$$\{q_{d,\bar{\varepsilon}}(\lambda) | \lambda \in \mathbb{R}_+^*\} \cap]-\infty, f_\infty(d)] \neq \varnothing. \tag{1.8}$$

Define $\varepsilon_d^* > 0$ as the supremum of all the $\bar{\varepsilon} > 0$ for which (1.8) holds. Then the situation can be fully described as follows [13, section II].

(a) ε lies in the interval $]0, \varepsilon_d^*[$; then
  (i) $f'(x_0; d) < f_\varepsilon'(x_0; d) < f_\infty(d)$;
  (ii) $\Lambda_d(\varepsilon)$ is a nonempty compact interval;
  (iii) $0 \notin M_d(\varepsilon)$.
(b) ε lies in the interval $]\varepsilon_d^*, +\infty[$; then
  (i) $f'(x_0; d) < f_\varepsilon'(x_0; d) = f_\infty(d)$;

(ii) $\Lambda_d(\varepsilon)$ is empty;

(iii) $M_d(\varepsilon) = \{0\}$.

(c) $\varepsilon = \varepsilon_d^*$; then

(i) $f'(x_0; d) < f'_{\varepsilon_d^*}(x_0; d) = f_\infty(d)$;

(ii) $\Lambda_d(\varepsilon_d^*)$, if nonempty, is unbounded; that holds if and only if there exists $\underline{\lambda} > 0$ such that

$$f_d(\lambda) = f(x_0) + \lambda f_\infty(d) - \varepsilon$$

for all $\lambda \geqslant \underline{\lambda}$.

It is convenient to extend the definition of ε_d^* to the first case we dealt with (f_d affine on $\mathbb{R}_+$) by making $\varepsilon_d^* = 0$. Moreover, it is easy to see that $\varepsilon_d^* < +\infty$ if and only if $f_\infty(d) < +\infty$.

In the sequel, we shall use $M_d(\varepsilon)$ and $\Lambda_d(\varepsilon)$ indifferently, bearing in mind their relationship [cf (1.6)]. The advantage of considering $M_d(\varepsilon)$ is that it is nonempty for all $\varepsilon > 0$ so that it can be used in global statements (i.e., valid for all $\varepsilon > 0$). Using $\Lambda_d(\varepsilon)$, whose definition is more conspicuous, will require $\varepsilon \in]0, \varepsilon_d^*[$ in most statements.

At those points x_0 where the function $v_{d,\varepsilon}: x \mapsto f'_\varepsilon(x; d)$ is not differentiable, it admits however a directional derivative $v'_{d,\varepsilon}(x_0; \delta)$ for all δ. The exact formulation of $v'_{d,\varepsilon}(x_0; \delta)$, which we also denote by $f''_\varepsilon(x_0; d, \delta)$, was worked out by Lemaréchal and Nurminskii [14] under the assumption that f is coercive (i.e., $\lim_{\|x\| \to +\infty} f(x)/\|x\| = +\infty$) and generalized by Auslender [1] to arbitrary finite functions. Here we shall only consider the case where $\delta = d$; the formula giving $f''_\varepsilon(x_0; d, d)$ comes as follows:

$$f''_\varepsilon(x_0; d, d) = \bar{\mu}_d(\varepsilon)[f'_\varepsilon(x_0; d) - f'(x_0; d)], \tag{1.9}$$

where $\bar{\mu}_d(\varepsilon)$ stands for $\min\{\mu | \mu \in M_d(\varepsilon)\}$. Following the classification we have recalled above, we retain that

(i) $f''_\varepsilon(x_0; d, d) = 0$ for all $\varepsilon \in [\![\varepsilon_d^*, +\infty[$;

(ii) $f''_\varepsilon(x_0; d, d) = [f'_\varepsilon(x_0; d) - f(x_0; d)]/\bar{\lambda}_d(\varepsilon)$ for all $\varepsilon \in]0, \varepsilon_d^*[$, with $\bar{\lambda}_d(\varepsilon) = \max\{\lambda | \lambda \in \Lambda_d(\varepsilon)\}$.

It is clear that the study of the qualitative and limiting behaviours of the function $\varepsilon \mapsto f''_\varepsilon(x_0; d, d)$ rely on the knowledge of the behaviours of the function $\varepsilon \mapsto f'_\varepsilon(x_0; d)$ and of the multifunction $\varepsilon \rightrightarrows \Lambda_d(\varepsilon)$. That will precisely be the aim in section 2.

2. BEHAVIOUR OF $f'_\varepsilon(x_0; d)$ AS A FUNCTION OF ε

Let θ_d be defined for all $\varepsilon \in \mathbb{R}$ as follows:

$$\theta_d(\varepsilon) = \begin{cases} f'_\varepsilon(x_0; d) \text{ if } \varepsilon \geqslant 0 \\ -\infty \text{ if } \varepsilon < 0. \end{cases}$$

As we shall see it in the sequel, this extension of the definition of $f'_\varepsilon(x_0; d)$ to the $\varepsilon < 0$ is not only convenient but also natural. θ_d is known to be an increasing function of ε satisfying

$$\lim_{\varepsilon \to 0^+} \theta_d(\varepsilon) = \theta_d(0) = f'(x_0; d).$$

The main features of the behaviour of θ_d (on $\mathbb{R}_+$) will be given in the next results. For that purpose, we recall that the functions $r_{d,\varepsilon}$, $\varepsilon > 0$, defined in the previous section are convex on

$\mathbb{R}_+$. We agree on extending the $r_{d,\varepsilon}$ to all of $\mathbb{R}$ by making $r_{d,\varepsilon}(\mu) = +\infty$ whenever $\mu < 0$. r_d and $-\theta_d$ are thus two lower-semicontinuous convex proper functions on $\mathbb{R}$ and the question of their relationship is answered now.

PROPOSITION 2.1.

$$(r_d)^*(\mu^*) = -\theta_d(-\mu^*) \tag{2.1}$$

for all $\mu^* \in \mathbb{R}$.

Proof. By definition,

$$(r_d)^*(\mu^*) = \sup_{\mu>0} \left\{ \mu\mu^* - \mu\left[f\left(x_0 + \frac{d}{\mu}\right) - f(x_0) \right] \right\} \tag{2.2}$$

for all $\mu^* \in \mathbb{R}$. Since $\lim_{\mu\to+\infty} r_d(\mu) = f'(x_0; d) < +\infty$, we have that $(r_d)^*(\mu^*) = +\infty$ whenever $\mu^* > 0$. Whence the equality (2.1) for $\mu^* > 0$. If $\mu^* \leqslant 0$, it readily comes from (2.2) that

$$\begin{aligned}(r_d)^*(\mu^*) &= -\inf_{\mu>0} \left\{ \mu\left[f\left(x_0 + \frac{d}{\mu}\right) - f(x_0) \right] + \mu(-\mu^*) \right\} \\ &= -f'_{(-\mu^*)}(x_0; d).\end{aligned}$$

Hence the announced result is proved. ■

Since $r_{d,\varepsilon}(\mu) = r_d(\mu) + \varepsilon\mu$ for all μ, the result of proposition 2.1 yields that

$$(r_{d,\varepsilon})^*(\mu^*) = -\theta_d(\varepsilon - \mu^*) \qquad \text{for all } \mu^*. \tag{2.3}$$

From its definition, one knows that θ_d is a concave function of ε. The segment where θ_d is actually strictly increasing can be delineated with the aid of its superdifferential $\partial\theta_d$. As it is clear from (2.1), the calculation of $\partial\theta_d(\varepsilon)$ for $\varepsilon \geqslant 0$ amounts to that of the subdifferential of r_d at $\mu \geqslant 0$.

PROPOSITION 2.2. We have that

(i) $\partial r_d(0) =]-\infty, -\varepsilon_d^*]]$

(ii) $\partial r_d(\mu) = -\dfrac{1}{\mu} < \partial f\left(x_0 + \dfrac{d}{\mu}\right), d > + f\left(x_0 + \dfrac{d}{\mu}\right) - f(x_0)$ for all $\mu > 0$.

Proof. $\partial r_d(0)$ is empty when $r_d(0) = f_\infty(d) = +\infty$. In such a case, we know that $\varepsilon_d^* = +\infty$. Assume that $f_\infty(d) < +\infty$; $\mu^* \in \partial r_d(0)$ if and only if

$$\mu\left[f\left(x_0 + \frac{d}{\mu}\right) - f(x_0) + (-\mu^*) \right] \geqslant f_\infty(d) \qquad \text{for all } \mu > 0. \tag{2.4}$$

According to what has been recalled in section 1, the above inequality holds if and only if $-\mu^* \geqslant \varepsilon_d^*$. Whence the expression of $\partial r_d(0)$. As for the expression of $\partial r_d(\mu)$, $\mu > 0$, it is just a matter of applying the chain rule already used for proposition 1.2. ■

THEOREM 2.3. We have that

$$\partial\theta_d(\varepsilon) = M_d(\varepsilon) \qquad \text{for all } \varepsilon \geqslant 0. \tag{2.5}$$

Furthermore, θ_d is strictly increasing on $[0, \varepsilon_d^*]]$ and takes a constant value $(=f_\infty(d))$ on $[\![\varepsilon_d^*, +\infty[$.

Proof. By definition of the superdifferential,

$$\partial\theta_d(\varepsilon) = -\partial(-\theta_d)(\varepsilon) \qquad \text{for all } \varepsilon \geqslant 0.$$

According to the relationship (2.1),

$$-\mu^* \in \partial(-\theta_d)(\varepsilon) \Leftrightarrow \mu^* \in \partial(r_d^*)(-\varepsilon) \Leftrightarrow -\varepsilon \in \partial r_d(\mu^*).$$

Therefore, it follows from proposition 2.2 that

$$0 \in \partial\theta_d(\varepsilon) \Leftrightarrow \varepsilon \geqslant \varepsilon_d^*$$

$$\left.\begin{array}{l}\mu^* \in \partial\theta_d(\varepsilon)\\ \mu^* > 0\end{array}\right\} \Leftrightarrow 0 \in -\frac{1}{\mu^*} < \partial f\left(x_0 + \frac{d}{\mu^*}\right), d > + f\left(x_0 + \frac{d}{\mu^*}\right) - f(x_0) + \varepsilon.$$

So, in all cases, we can rephrase the above as

$$\mu^* \in \partial\theta_d(\varepsilon) \Leftrightarrow \mu^* \text{ minimizes } r_{d,\varepsilon}(\mu) \text{ on } \mathbb{R}_+.$$

Hence $\partial\theta_d(\varepsilon) = M_d(\varepsilon)$ for all $\varepsilon \geqslant 0$.

We have shown that $0 \notin M_d(\varepsilon)$ whenever $0 < \varepsilon < \varepsilon_d^*$. Thus, θ_d is strictly monotone on $[0, \varepsilon_d^*]]$. ■

Remark 1. since $\varepsilon \rightrightarrows M_d(\varepsilon)$ is a superdifferential mapping, $M_d(\varepsilon)$ is reduced to $\{\mu_d(\varepsilon)\}$ except at countably many of the ε. In other words, since $\Lambda_d(\varepsilon)$ is empty for all $\varepsilon > \varepsilon_d^*$ and $\Lambda_d(\varepsilon) = 1/M_d(\varepsilon)$ for all $\varepsilon \in]0, \varepsilon_d^*[$, $q_{d,\varepsilon}$ achieves its minimum value on a single point $\lambda_d(\varepsilon)$ except maybe on an at most countable set of $\varepsilon \in]0, \varepsilon_d^*[$. All the properties of the multifunction $\varepsilon \rightrightarrows M_d(\varepsilon)$ on $\mathbb{R}_+$ (and those of $\varepsilon \rightrightarrows \Lambda_d(\varepsilon)$ on $]0, \varepsilon_d^*[$) come from the fact that $\{(\varepsilon, \mu) | \varepsilon \geqslant 0, \mu \in M_d(\varepsilon)\}$ is a "complete non-increasing curve" in $\mathbb{R}^2$ [16, section 24].

Remark 2. The knowledge of $\Lambda_d(\varepsilon)$ and that of q_d (or equivalently r_d) on $\mathbb{R}_+$ allows us to calculate $f_\varepsilon'(x_0; d)$ since, by definition,

$$[f(x_0 + \lambda d) - f(x_0) + \varepsilon]\lambda^{-1} = f_\varepsilon'(x_0; d) \text{ for all } \lambda \in \Lambda_d(\varepsilon).$$

It is noteworthy that the converse is also true. If $f_\varepsilon'(x_0; d)$ is known for all $\varepsilon > 0$, the relation (2.1) yields the expression of r_d on $\mathbb{R}_+$, namely

$$r_d(\mu) = \sup_{\varepsilon > 0} [f_\varepsilon'(x_0; d) - \varepsilon\mu] \qquad \text{for all } \mu \geqslant 0.\ddagger \tag{2.6}$$

As for $\Lambda_d(\varepsilon)$, it depends on the behaviour of $f_\varepsilon'(x_0; d)$ around a given ε. For example, if $\varepsilon \mapsto f_\varepsilon'(x_0; d)$ is differentiable at $\varepsilon > 0$ with a derivative $\mu_d(\varepsilon)$, that means that either $\Lambda_d(\varepsilon)$ is empty (when $\mu_d(\varepsilon) = 0$) or $\Lambda_d(\varepsilon) = \{1/\mu_d(\varepsilon)\}$ (whenever $\mu_d(\varepsilon) > 0$).

‡ Lemaréchal gets at the same result by showing that for all $\lambda > 0$ there exists $\varepsilon(\lambda) \geqslant 0$ such that $q_d(\lambda) = f_{\varepsilon(\lambda)}'(x_0; d)$.

Note also that, in an integral representation, we have the following:

$$\begin{aligned} f'_\varepsilon(x_0; d) &= f'(x_0; d) + \int_0^\varepsilon M_d(\alpha)\,\mathrm{d}\alpha \qquad \text{for all } \varepsilon \geqslant 0, \\ &= f'(x_0; d) + \int_0^\varepsilon \frac{\mathrm{d}\alpha}{\Lambda_d(\alpha)} \qquad \text{for all } \varepsilon < \varepsilon_d^*. \end{aligned} \tag{2.7}$$

The result of theorem 2.3 contains all the information on the behaviour of $f'_\varepsilon(x_0; d)$ when $\varepsilon \to +\infty$ and when $\varepsilon \to 0^+$. These limiting behaviours will be discussed later. Beforehand, there is a question which arises naturally from the foregoing results. Since $\varepsilon \rightrightarrows \Lambda_d(\varepsilon)$ is a subdifferential mapping, there exists a convex function, uniquely determined up to an additive constant, whose subdifferential is precisely $\Lambda_d(\varepsilon)$ for all $\varepsilon \in]0, \varepsilon_d^*[$. The question however is to display a convex function φ_d, directly derived from $f'_\varepsilon(x_0; d)$, whose subdifferential is $\Lambda_d \circ \sigma$, for a certain homeomorphism σ. The answer relies upon the inverse function theorem for locally Lipschitz functions [5]. Since only the one-dimensional case is needed here, we recall the results peculiar to that setting [11].

THEOREM 2.4. Let f be a locally Lipschitz function on $]a, b[$ and $x_0 \in]a, b[$. If $0 \notin \partial f(x_0)$, f is strictly monotone in a neighbourhood of x_0 and the inverse function f^{-1} is Lipschitz in a neighbourhood of $f(x_0)$ with

$$\partial f^{-1}(f(x_0)) = \frac{1}{\partial f(x_0)}.$$

In such a case, f is said to be a local lipeomorphism at x_0.

COROLLARY 2.5. Let f be a locally Lipschitz function on $]a, b[$ such that $0 \notin \partial f(x)$ for all $x \in]a, b[$. Then f^{-1} is locally Lipschitz on $]\alpha, \beta[= f(]a, b[)$ and

$$\partial f^{-1} = \frac{1}{\partial f \circ f^{-1}}.$$

Let now φ_d be the function defined on $\mathbb{R}$ as follows:

$$\varphi_d(x) = \begin{cases} 0 \text{ if } c \leqslant f'(x_0; d); \\ \text{the unique } \varepsilon \in]0, \varepsilon_d^*[\text{ for which } f'_\varepsilon(x_0; d) = c, \text{ when} \\ f'(x_0; d) < c < f_\infty(d); \\ \left.\begin{array}{l} \varepsilon_d^* \text{ if } c = f_\infty(d); \\ +\infty \text{ if } c > f_\infty(d). \end{array}\right\} \text{ holds only if } f_\infty(d) < +\infty. \end{cases}$$

From the geometric viewpoint, φ_d is the function whose epigraph is the image of the hypograph of θ_d under the transformation $(u, v) \leftrightarrow (v, u)$.

THEOREM 2.6. φ_d is a lower-semicontinuous convex function, strictly increasing on $[f'(x_0; d), f_\infty(d)]]$, and whose subdifferential is described as:

$$\partial\varphi_d(c) = \begin{cases} [0, a_d^*]] \text{ if } c = f'(x_0; d); \\ \Lambda_d(\varepsilon) \text{ if } c = f_\varepsilon'(x_0; d) \in]f'(x_0; d), f_\infty(d)[; \\ \Lambda_d(\varepsilon_d^*) \text{ if } c = f_\infty(d) < +\infty \text{ and } \varepsilon_d^* > 0. \end{cases}$$

Proof. When $c \in]f'(x_0; d), f_\infty(d)[$, the calculation of $\partial\varphi_d(c)$ is just a matter of applying corollary 2.5. As for the end-points, the description of the subdifferential of φ_d derives from the definition itself. ■

Beside the qualitative behaviour of $\varepsilon \mapsto f_\varepsilon'(x_0; d)$ on $\mathbb{R}_+$ which we have studied, we are now interested in the limiting behaviour of $[f_\varepsilon'(x_0; d) - f'(x_0; d)]/\varepsilon$ when $\varepsilon \to +\infty$ or $\varepsilon \to 0^+$.

Consider first $\varepsilon \to +\infty$. If $f_\infty(d) < +\infty$, we know that $f_\varepsilon'(x_0; d) = f_\infty(d)$ for $\varepsilon \geqslant \varepsilon_d^*$. When $f_\infty(d) = +\infty$, we have that $\lim_{\varepsilon \to +\infty} f_\varepsilon'(x_0; d) = +\infty$; one might be tempted to describe the limiting behaviour of $f_\varepsilon'(x_0; d)$ as equivalent to that of $k_1 + k_2\varepsilon$ for some $k_1, k_2 > 0$. The next result shows there is nothing of the kind.

THEOREM 2.7. We have that

(i) $$\lim_{\varepsilon \to +\infty} \frac{f_\varepsilon'(x_0; d)}{\varepsilon} = 0; \tag{2.8}$$

(ii) $$\lim_{\varepsilon \to +\infty} M_d(\varepsilon) = \{0\}. \tag{2.9}$$

Proof. The recession function of $(r_d)^*$ is known to be the support function of the domain of r_d [16, theorem 13.3]. Therefore, it follows from (2.1) that

$$\lim_{\varepsilon \to +\infty} \frac{-\theta_d(\varepsilon) + f'(x_0; d)}{\varepsilon} = 0.$$

whence the result (2.8) is proved.

Since $M_d(\varepsilon)$ is the superdifferential of θ_d at $\varepsilon > 0$ (theorem 2.3), we have that

$$f_\zeta'(x_0; d) \leqslant f_\varepsilon'(x_0; d) + \mu_d(\varepsilon)(\zeta - \varepsilon) \quad \text{for all } \zeta > 0, \varepsilon > 0, \quad \text{and } \mu_d(\varepsilon) \in M_d(\varepsilon).$$

by making $\zeta = \varepsilon/2$ in the above, we have that

$$\mu_d(\varepsilon) + 2\frac{f_{\varepsilon/2}'(x_0; d)}{\varepsilon} \leqslant 2\frac{f_\varepsilon'(x_0; d)}{\varepsilon}.$$

Observing that

$$\frac{f'(x_0; d)}{\varepsilon} \leqslant \frac{f_{\varepsilon/2}'(x_0; d)}{\varepsilon} \leqslant \frac{f_\varepsilon'; x_0; d)}{\varepsilon} \quad \text{and } \mu_d(\varepsilon) \geqslant 0 \quad \text{for all } \varepsilon > 0,$$

we deduce from the first limiting result (2.8) that

$$\lim_{\varepsilon \to +\infty} \mu_d(\varepsilon) = 0. \quad ■$$

The limiting behaviour of $[f'_\varepsilon(x_0;d) - f'(x_0;d)]/\varepsilon$ when $\varepsilon \to 0^+$ is far more important to tackle. The result is actually contained in the fact that $\partial\theta_d(0) = M_d(0)$ (theorem 2.3) and the description of $M_d(0)$ as $[\![1/a_d^*, +\infty[$ [cf. relation (1.5)].

THEOREM 2.8. We have that

(i) $$\lim_{\varepsilon\to 0^+} \frac{f'_\varepsilon(x_0;d) - f'(x_0;d)}{\varepsilon} = \frac{1}{a_d^*} \text{ in } \bar{\mathbb{R}}_+; \tag{2.10}$$

(ii) $$\lim_{\varepsilon\to 0^+} \Lambda_d(\varepsilon) = \{a_d^*\} \text{ whenever } a_d^* < +\infty. \tag{2.11}$$

Proof. By the mere fact that $(\theta_d)'_+(0) = 1/a_d^*$ and $\theta_d(0) = f'(x_0;d)$ we get the result (2.10).

As to the limit of $\Lambda_d(\varepsilon)$ when $\varepsilon \to 0^+$, the result can be viewed as a simple consequence of theorem 2.6. ■

Remark 1. When $0 < a_d^* < +\infty$, we infer from (2.10) that

$$f'_\varepsilon(x_0;d) = f'(x_0;d) + \frac{\varepsilon}{a_d^*} + o(\varepsilon) \qquad \text{when } \varepsilon \to 0^+. \tag{2.12}$$

This situation typically occurs when f_d is polyhedral (but not affine) on $\mathbb{R}_+$. If it is so, the function $\varepsilon \mapsto f'_\varepsilon(x_0;d)$ is itself polyhedral on $\mathbb{R}_+$ and the term $o(\varepsilon)$ in (2.12) is null for ε small enough.

Remark 2. The case $a_d^* = 0$ is of particular importance since it occurs "the most often". We recall that $a_d^* = 0$ is equivalent to assuming that

$$[f(x_0 + \lambda d) - f(x_0)]\lambda^{-1} > f'(x_0;d) \qquad \text{for all } \lambda > 0.$$

So, according to theorem 2.8, the above assumption holds if and only if ($\varepsilon_d^* > 0$ and) all the $\lambda_d(\varepsilon) \in \Lambda_d(\varepsilon)$ converge to 0 as $\varepsilon \to 0^+$. This result was proved by the author [13, corollary 4.4] in a somewhat different way.

Due to the result (2.11), the expansion (2.12) is also equivalent to

$$f'_\varepsilon(x_0;d) = f'(x_0;d) + \frac{\varepsilon}{\lambda_d(\varepsilon)} + o(\varepsilon) \qquad \text{when } \varepsilon \to 0^+, \tag{2.13}$$

where $\lambda_d(\varepsilon)$ is arbitrarily chosen in $\Lambda_d(\varepsilon)$. When $a_d^* = 0$, both quantities $[f'_\varepsilon(x_0;d) - f'(x_0;d)]/\varepsilon$ and $1/\lambda_d(\varepsilon)$ go to $+\infty$ when $\varepsilon \to 0^+$ and their difference is not secured to converge to 0. Therefore, as a general rule, a result like (2.13) does not hold in such a case. Counter-examples will be given later. Nevertheless, we have the following weaker result.

COROLLARY 2.9. Assume that $a_d^* < +\infty$. Then

$$f'_\varepsilon(x_0;d) - f'(x_0;d) - \frac{\varepsilon}{\lambda_d(\varepsilon)} \to 0^+ \qquad \text{when } \varepsilon \to 0^+, \tag{2.14}$$

whatever $\lambda_d(\varepsilon) \in \Lambda_d(\varepsilon)$.

Proof. Since $1/\lambda_d(\varepsilon)$ is a supergradient of θ_d at $\varepsilon > 0$ (theorem 2.3), we have that

$$f'_\varepsilon(x_0; d) - f'(x_0; d) \geqslant \frac{\varepsilon}{\lambda_d(\varepsilon)} \qquad \text{for all } \varepsilon \in]0, \varepsilon_d^*[. \tag{2.15}$$

As a result, $\varepsilon/\lambda_d(\varepsilon)$ converges to 0^+ with ε and the announced result is secured. ■

The next example illustrates the 'optimality' of the foregoing results.

Example. Let f be defined on $\mathbb{R}$ by $f(x) = |x|^\alpha$ with $\alpha > 1$. For $x_0 = 0$ and $d = 1$, we have that

$$\Lambda_d(\varepsilon) = \left\{\left(\frac{\varepsilon}{\alpha - 1}\right)^{1/\alpha}\right\}, \; f'_\varepsilon(x_0; d) = \alpha\left(\frac{\varepsilon}{\alpha - 1}\right)^{(\alpha-1)/\alpha} \qquad \text{for all } \varepsilon > 0.$$

As indicated in corollary 2.9, $\varepsilon/\lambda_d(\varepsilon)$ converges to 0^+ with ε. Without further information on the behaviour of f_d near 0^+, this convergence result cannot be sharpened to $\varepsilon/[\lambda_d(\varepsilon)]^\beta$ for some $\beta > 1$. Indeed, in the afore-mentioned case, we observe that

$$\lim_{\varepsilon \to 0^+} \frac{\varepsilon}{[\lambda_d(\varepsilon)]^\beta} = +\infty \text{ whenever } \beta > \alpha.$$

For the same example, we have that

$$\frac{f'_\varepsilon(x_0, d) - f'(x_0; d)}{\varepsilon} - \frac{1}{\lambda_d(\varepsilon)} \to +\infty \qquad \text{when } \varepsilon \to 0^+.$$

Remark. Since $f'_\varepsilon(x_0; d) - f'(x_0; d)$ converges to 0^+ with ε, one might be tempted to try to find $\sigma > 0$ such that

$$f'_\varepsilon(x_0; d) - f'(x_0; d) = o(\varepsilon^\sigma) \quad \text{when } \varepsilon \to 0^+. \tag{2.16}$$

As seen earlier, (2.16) holds true with $\sigma = 1$ whenever $a_d^* > 0$. But, as a general rule, an estimate like (2.16) is hopeless when $a_d^* = 0$; the above-studied example with α approaching 1^+ clearly shows it. More will be said about such asymptotic estimates in section 3 [cf. relation (3.10)].

3. BEHAVIOUR OF $f''_\varepsilon(x_0; d, d)$ AS A FUNCTION OF ε

We henceforth cast aside the trivial case where $a_d^* = +\infty$.

Given $\varepsilon \in]0, \varepsilon_d^*[$, we know from section 1 that

$$f''_\varepsilon(x_0; d, d) = [f'_\varepsilon(x_0; d) - f'(x_0; d)]/\bar{\lambda}_d(\varepsilon),$$

where $\bar{\lambda}_d(\varepsilon)$ stands for $\max\{\lambda | \lambda \in \Lambda_d(\varepsilon)\}$. It might be convenient to denote by ρ_d the function $\varepsilon \mapsto f''_\varepsilon(x_0; d, d)$. According to what has been seen earlier, ρ_d is the quotient of two decreasing functions of ε and

$$\rho_d(\varepsilon) \geqslant \varepsilon/[\bar{\lambda}_d(\varepsilon)]^2 \qquad \text{for all } \varepsilon \in]0, \varepsilon_d^*[. \tag{3.1}$$

The behaviour of $\rho_d(\varepsilon)$ when $\varepsilon \to 0^+$ can be wild and it is not even secured that

$$\limsup_{\varepsilon \to 0^+} \rho_d(\varepsilon) < +\infty.$$

Likewise, ρ_d is not expected to be increasing or decreasing on $]0, \varepsilon_d^*[$. We begin by focusing our attention on the limiting behaviour of $\rho_d(\varepsilon) = f_\varepsilon''(x_0; d, d)$ when $\varepsilon \to 0^+$.

There is a case which can be settled at once, namely when $a_d^* > 0$. In such a case, since $\bar{\lambda}_d(\varepsilon)$ converges to a_d^*, we infer from (2.12) that

$$f_\varepsilon''(x_0; d, d) = \frac{\varepsilon}{(a_d^*)^2} + o(\varepsilon) \qquad \text{when } \varepsilon \to 0^+. \tag{3.2}$$

Likewise, the counterpart of the expansion (2.13) is that

$$f_\varepsilon''(x_0; d, d) = \frac{\varepsilon}{[\lambda_d(\varepsilon)]^2} + o(\varepsilon) \qquad \text{when } \varepsilon \to 0^+, \tag{3.3}$$

whatever $\lambda_d(\varepsilon) \in \Lambda_d(\varepsilon)$.

Let now $a_d^* = 0$. A result like (3.3) is hopeless in such a situation; even a weaker result, namely the existence of

$$\lim_{\varepsilon \to 0^+} \left[f_\varepsilon''(x_0; d, d) - \frac{\varepsilon}{[\bar{\lambda}_d(\varepsilon)]^2} \right] \qquad \text{in } \mathbb{R}_+ \tag{3.4}$$

is not generally secured. To convince oneself of that, consider the function f in the example displayed at the end of section 2. We have that

$$f_\varepsilon''(x_0; d, d) - \frac{\varepsilon}{[\lambda_d(\varepsilon)]^2} = (\alpha - 1)^{(\alpha+2)/\alpha} \varepsilon^{(\alpha-2)/\alpha}.$$

The 'critical value' for securing a result like (3.4) is clearly $\alpha = 2$, that is, precisely the minimal value of α ensuring the twice-differentiability of f. So, we are led to make assumptions on the behaviour of f_d on $\mathbb{R}_+$. We shall say that f has a second right Dini derivative at x_0 in the d direction if

$$D_r'' f(x_0; d) = \lim_{\lambda \to 0^+} \frac{1}{\lambda} [f'(x_0 + \lambda d; d) - f'(x_0; d)] \tag{3.5}$$

exists and is finite. Likewise, f is said to have a second right de la Vallée–Poussin derivative at x_0 in the d direction if

$$V_r'' f(x_0; d) = \lim_{\lambda \to 0^+} \frac{1}{\lambda} \left[\frac{f(x_0 + \lambda d) - f(x_0)}{\lambda} - f'(x_0; d) \right] \tag{3.6}$$

exists and is finite. As a general rule, the existence of $D_r'' f(x_0; d)$ implies the existence of $V_r'' f(x_0; d)$ with $D_r'' f(x_0; d) = 2V_r'' f(x_0; d)$. However, for convex functions, the existence of $D_r'' f(x_0; d)$ and that of $V_r'' f(x_0; d)$ are equivalent. According to Busemann [3, p. 10], this result dates back to Jessen (1929). Jessen's result was stated under the assumption that f_d is differentiable at 0 but there is not trouble in getting rid of this requirement.

Note that, since $\lambda \mapsto f'(x_0 + \lambda d; d)$ is monotone on $\mathbb{R}_+$, the limit remains unchanged in (3.5) if one puts any element of $\langle \partial f(x_0 + \lambda d), d \rangle$ instead of $f'(x_0 + \lambda d; d)$. Hence, we can write symbolically

$$D_r'' f(x_0; d) = \lim_{\lambda \to 0^+} \frac{1}{\lambda} [\langle \partial f(x_0 + \lambda d), d \rangle - f'(x_0; d)]. \tag{3.7}$$

If one of the above-mentioned limits exists (in $\mathbb{R}_+$), we shall simply say that f has a second right derivative at x_0 in the direction d.

THEOREM 3.1. Assume that f has a second right derivative at x_0 in the d direction. Then

(i) $\lim_{\varepsilon \to 0^+} f''_\varepsilon(x_0; d, d) = D''_r f(x_0; d)$;

(ii) $\lim_{\varepsilon \to 0^+} \dfrac{\varepsilon}{[\lambda_d(\varepsilon)]^2} = \frac{1}{2} D''_r f(x_0; d) = V''_r f(x_0; d)$ for all $\lambda_d(\varepsilon) \in \Lambda_d(\varepsilon)$.

Proof. We consider only the case where $a_d^* = 0$ so that $\bar{\lambda}_d(\varepsilon)$ converges to 0^+ with ε.

(i) According to lemma 3.9 in [13], we have that

$$f'_\varepsilon(x_0; d) \in \langle \partial f(x_0 + \bar{\lambda}_d(\varepsilon) d), d \rangle.$$

Thus,

$$f''_\varepsilon(x_0; d, d) \in \frac{\langle \partial f(x_0 + \bar{\lambda}_d(\varepsilon) d), d \rangle - f'(x_0; d)}{\bar{\lambda}_d(\varepsilon)}$$

and the result on the limit of $f''_\varepsilon(x_0; d, d)$ derives from (3.7).

(ii) Due to the expressions of $f'_\varepsilon(x_0; d)$ and $f''_\varepsilon(x_0; d, d)$, we have that

$$\frac{1}{\bar{\lambda}_d(\varepsilon)} \left[\frac{f(x_0 + \bar{\lambda}_d(\varepsilon) d) - f(x_0)}{\bar{\lambda}_d(\varepsilon)} - f'(x_0; d) \right] = f''_\varepsilon(x_0; d, d) - \frac{\varepsilon}{[\bar{\lambda}_d(\varepsilon)]^2}.$$

The left-hand side of this inequality converges to $V''_r f(x_0; d) = \frac{1}{2} D''_r f(x_0; d)$, while $f''_\varepsilon(x_0; d, d)$ is proved to converge to $D''_r f(x_0; d)$. Hence,

$$\lim_{\varepsilon \to 0^+} \frac{\varepsilon}{[\bar{\lambda}_d(\varepsilon)]^2} = V''_r f(x_0; d) = \tfrac{1}{2} D''_r f(x_0; d).$$

Now, since $\varepsilon \rightrightarrows \Lambda_d(\varepsilon)$ is monotone on $]0, \varepsilon_d^*[$ (theorem 2.6), the above result holds for any $\lambda_d(\varepsilon) \in \Lambda_d(\varepsilon)$. ■

Remark 1. Assuming that $D''_r f(x_0; d)$ is non-null, we deduce from the results of theorem 3.1 that

$$\frac{f'_\varepsilon(x_0; d) - f'(x_0; d)}{[2\varepsilon D''_r f(x_0; d)]^{1/2}} \to 1 \text{ when } \varepsilon \to 0^+. \tag{3.8}$$

Hence, in such a case,

$$f'_\varepsilon(x_0; d) = f'(x_0; d) + [2\varepsilon D''_r f(x_0; d)]^{1/2} + o(\varepsilon^{1/2}) \tag{3.9}$$

and

$$f'_\varepsilon(x_0; d) - f'(x_0; d) = 0(\varepsilon^{1/2}). \tag{3.10}$$

A typical illustration of such relationships is, of course, that of quadratic functions. Let f be defined on $\mathbb{R}^n$ as

$$f(x) = \tfrac{1}{2} \langle Ax, x \rangle + \langle b, x \rangle + c,$$

where A is a symmetric positive-definite $n \times n$ matrix, b a vector in $\mathbb{R}^n$ and c a real number. Given x_0 and a non-null d, it is easy to check that

$$f'_\varepsilon(x_0; d) = f'(x_0; d) + (2\varepsilon\langle Ad, d\rangle)^{1/2},$$

$$\lambda_d(\varepsilon) = \left(\frac{2\varepsilon}{\langle Ad, d\rangle}\right)^{1/2}.$$

Remark 2. When $a_d^* > 0$, it is clear that f has a second right derivative in the d direction at the considered point x_0, namely $D''_r f(x_0; d) = 0$. So the setting of theorem 3.1 covers all the cases where $a_d^* > 0$. However, in such cases, asymptotic estimates sharper than those of (3.9) or (3.10) can be obtained [cf. (2.12) or (2.13)].

Remark 3. To require that f has a second right derivative at x_0 in the d direction is a milder assumption than might appear at first glance. Actually, this requirement is secured at almost all x_0 in all directions d. These x_0 are precisely those where ∂f is differentiable (Mignot's formulation) or, equivalently, those where f has a second order Taylor expansion (Alexandroff's formulation); for details we refer to [13, section IV.2] where further results on the limiting behaviour of $f''_\varepsilon(x; d, d)$ when $x \to x_0$ and $\varepsilon \to 0^+$ are displayed.

To pursue our argument, here is an example where f has a second right derivative at x_0 in all directions d without being differentiable at x_0. Let f_i, $i = 1, \ldots, m$, be convex twice continuously differentiable functions and set

$$f(x) = \max\{f_i(x) \mid i = 1, \ldots, m\}.$$

Demjanov and Malozemov [7, chapter III] proved that the expansion

$$f(x_0 + \lambda d) = f(x_0) + \lambda f'(x_0; d) + \frac{\lambda^2}{2} D''_r f(x_0; d) + o(\lambda^2)$$

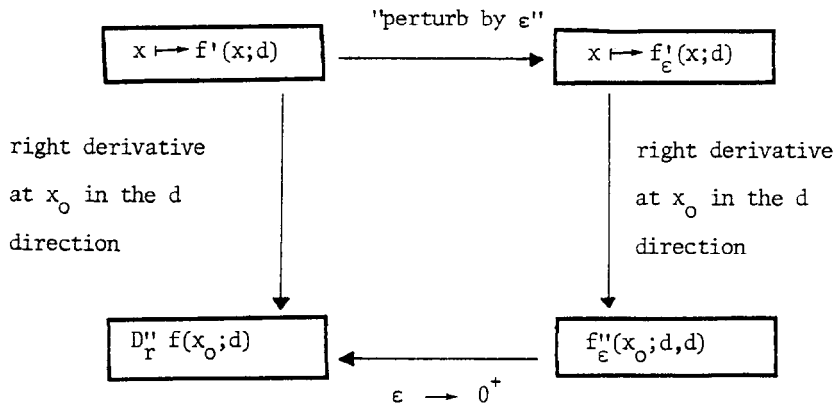

Fig. 1.

holds true with

$$D''_r f(x_0; d) = \max\{\langle \nabla^2 f_i(x_0)d, d\rangle | i \in I(x_0; d)\}$$
$$I(x_0; d) = \{i | f_i(x_0) = f(x_0) \text{ and } f'(x_0; d) = \langle \nabla f_i(x_0), d\rangle\}.$$

For such an f, the exact evaluations of $f'_\varepsilon(x_0; d)$, $\Lambda_d(\varepsilon)$ and $f''_\varepsilon(x_0; d, d)$ can be hard to carry out. Nevertheless, the asymptotic estimates displayed in remark 1 conspicuously show what to expect when ε is small enough.

To summarize, the results of theorem 3.1 can be schematized as shown in Fig. 1.

We now turn our attention to the following problem: how does $\rho_d(\varepsilon) = f''_\varepsilon(x_0; d, d)$ behave with ε on $]0, \varepsilon_d^*[$? For that purpose, it would be helpful to know the (sign of the) derivative $\rho'_d(\varepsilon)$ whenever it exists. Since ρ_d is the quotient of two decreasing functions of ε, namely

$$\rho_d(\varepsilon) = \frac{\theta_d(\varepsilon) - f'(x_0; d)}{\bar{\lambda}_d(\varepsilon)},$$

it is differentiable almost everywhere (a.e) on $]0, \varepsilon_d^*[$. We know that the (concave) function θ_d is differentiable at ε if and only if $\Lambda_d(\varepsilon)$ is reduced to a single element, say $\Lambda_d(\varepsilon) = \{\lambda_d(\varepsilon)\}$, and $\theta'_d(\varepsilon) = 1/\lambda_d(\varepsilon)$ (theorem 2.3). Likewise, $\Lambda_d(\varepsilon)$ is necessarily single-valued at a point ε where $\varepsilon \mapsto \bar{\lambda}_d(\varepsilon)$ [or, equivalently, $\varepsilon \rightrightarrows \Lambda_d(\varepsilon)$] is differentiable. Combining these facts together, we get the following.

THEOREM 3.2. Let $\varepsilon \in]0, \varepsilon_d^*[$ be a point where $\varepsilon \rightrightarrows \Lambda_d(\varepsilon)$ is differentiable, with $\lambda'_d(\varepsilon)$ as derivative. Then ρ_d is differentiable at ε with

$$\rho'_d(\varepsilon) = \frac{1 - [f'_\varepsilon(x_0; d) - f'(x_0; d)]\lambda'_d(\varepsilon)}{[\lambda_d(\varepsilon)]^2}. \tag{3.11}$$

This is of course a partial information since, as a general rule, the knowledge of $\rho'_d(\varepsilon)$ a.e. does not allow us to recover ρ_d from it. Additional assumptions have to be made to secure, for example, that ρ_d is locally Lipschitz on $]0, \varepsilon_d^*[$.

Since $\varepsilon \rightrightarrows \Lambda_d(\varepsilon)$ is a subdifferential mapping (theorem 2.6), to assume that $\Lambda_d(\varepsilon) = \{\lambda_d(\varepsilon)\}$ for all $\varepsilon \in]0, \varepsilon_d^*[$ is sufficient to ensure that λ_d and thus ρ_d are continuous on $]0, \varepsilon_d^*[$. To get more, namely that λ_d (as also ρ_d) is locally Lipschitz, we must assume more on the behaviour of f on the half-line $x_0 + \mathbb{R}_+ d$.

We shall say that f_d is $C^{1,1}$ on $\mathbb{R}_+^*$ if f_d is differentiable on $\mathbb{R}_+^*$ and if its derivative $(f_d)'$ is locally Lipschitz. We shall denote by $\partial^2 f_d(\lambda)$ the generalized derivative (in Clarke's sense) of $(f_d)'$ at $\lambda > 0$ [4], that is the smallest interval in $\mathbb{R}$ containing $\{\lambda^* | \exists \lambda_i \to \lambda, f_d$ twice differentiable at λ_i and $(f_d)''(\lambda_i) \to \lambda^*\}$. $\partial^2 f_d(\lambda)$ can be thought as a generalized second derivative of f_d at λ and clearly $\partial^2 f_d(\lambda) \subset \mathbb{R}_+$. Note that $\partial^2 f_d(\lambda) = \{(f_d)''(\lambda)\}$ for all $\lambda > 0$ whenever f_d is C^2 on $\mathbb{R}_+^*$.

THEOREM 3.3. Assume that f_d is $C^{1,1}$ on $\mathbb{R}_+^*$ with $\partial^2 f_d(\lambda) \subset \mathbb{R}_+^*$ for all $\lambda > 0$. Then

(a) $\Lambda_d(\varepsilon) = \{\lambda_d(\varepsilon)\}$ for all $\varepsilon \in]0, \varepsilon_d^*[$;

(b) $\lambda_d :]0, \varepsilon_d^*[\to \mathbb{R}_+^*$ is a strictly increasing, surjective, locally Lipschitz mapping whose generalized derivative is given as

$$\partial\lambda_d(\varepsilon) = \frac{1}{\lambda_d(\varepsilon)\partial^2 f_d(\lambda_d(\varepsilon))} \quad \text{for all } \varepsilon \in]0, \varepsilon_d^*[. \tag{3.12}$$

Proof. Clearly f_d is strictly convex on $\mathbb{R}_+$. As a result, $a_d^* = 0$ and $\varepsilon_d^* > 0$. According to proposition 1.2, given $\varepsilon \in]0, \varepsilon_d^*[$, $\lambda \in \Lambda_d(\varepsilon)$ if and only if

$$\varphi(\varepsilon, \lambda) = -\lambda(f_d)'(\lambda) + f_d(\lambda) - f_d(0) + \varepsilon = 0. \tag{3.13}$$

Now, since the generalized partial derivative $\hat{\partial}_\lambda\varphi(\varepsilon, \lambda)$ of φ with respect to λ is $-\lambda\partial^2 f_d(\lambda)$, it is a matter of applying the implicit function theorem for locally Lipschitz mappings (cf. [5], [9]) to get that the unique $\lambda_d(\varepsilon)$ of $\Lambda_d(\varepsilon)$ defines a locally Lipschitz mapping λ_d. On the one hand, because $a_d^* = 0$, we know that $\lim_{\varepsilon \to 0^+} \lambda_d(\varepsilon) = 0$ [cf. relation (2.11)]. On the other hand, because f_d is not affine for λ large enough, we have that $\lim_{\varepsilon \to \varepsilon_d^*} \lambda_d(\varepsilon) = +\infty$. Thus, λ_d is an increasing surjective mapping from $]0, \varepsilon_d^*[$ into $\mathbb{R}_+^*$§.

To calculate the generalized gradient $\partial\lambda_d(\varepsilon)$ exactly, we go back to the way used for proving the implicit function theorem [9, pp. 95–96]. Let $\varepsilon_0 \in]0, \varepsilon_d^*[$, $\lambda_0 = \lambda_d(\varepsilon_0)$, and g be defined in a neighbourhood of $(\varepsilon_0, \lambda_0)$ by

$$g(\varepsilon, \lambda) = (\varepsilon, \varphi(\varepsilon, \lambda)).$$

For $(\varepsilon_i, \lambda_i)$ in a neighbourhood of $(\varepsilon_0, \lambda_0)$, g is differentiable at $(\varepsilon_i, \lambda_i)$ if and only if $(f_d)'$ is differentiable at λ_i. The Jacobian matrix of g at such an $(\varepsilon_i, \lambda_i)$ is described as follows:

$$Jg(\varepsilon_i, \lambda_i) = \begin{bmatrix} 1 & 0 \\ 1 & -\lambda_i(f_d)''(\lambda_i) \end{bmatrix}. \tag{3.14}$$

There exist an open neighbourhood V of $(\varepsilon_0, \lambda_0)$, an open neighbourhood W_1 of $g(\varepsilon_0, \lambda_0) = (\varepsilon_0, 0)$, such that g is a lipeomorphism of V on W_1. Let h be the inverse lipeomorphism; it is of the form:

$$h(\varepsilon, z) = (\varepsilon, k(\varepsilon, z)), (\varepsilon, z) \in W_1. \tag{3.15}$$

The following is easy to prove: if g is differentiable at a point $(\varepsilon_i, \lambda_i)$, and $Jg(\varepsilon_i, \lambda_i)$ is its Jacobian matrix, then h is differentiable at $g(\varepsilon_i, \lambda_i)$ and its Jacobian matrix agrees with $[Jg(\varepsilon_i, \lambda_i)]^{-1}$. As a result, the generalized Jacobian matrix of h at $(\varepsilon_0, 0)$ is the set

$$\mathcal{J}h(\varepsilon_0, 0) = \operatorname{co}\left\{\lim_{i \to +\infty} [Jg(\varepsilon_i, \lambda_i)]^{-1}\right\},$$

where $(\varepsilon_i, \lambda_i)$ converges to $(\varepsilon_0, \lambda_0)$ and g is differentiable at $(\varepsilon_i, \lambda_i)$. Therefore, it comes from

§ In general, we simply have that $\Lambda_d([0, \varepsilon_d^*]) = \mathbb{R}_+^*$.

(3.14) that

$$\mathcal{T}h(\varepsilon_0, 0) = \left\{ \begin{bmatrix} 1 & 0 \\ x^* & -x^* \end{bmatrix}; x^* \in \frac{1}{\lambda_0 \partial^2 f_d(\lambda_0)} \right\}. \tag{3.16}$$

Now, since h is the inverse function of g, the following assertions are equivalent:

(i) $\quad (\varepsilon, \lambda) \in V \quad$ and $\quad z = \varphi(\varepsilon, \lambda)$;

(ii) $\quad (\varepsilon, z) \in W_1 \quad$ and $\quad \lambda = k(\varepsilon, z)$.

By posing $W = \{\varepsilon | (\varepsilon, 0) \in W_1\}$, we infer from the above that

$$(\varepsilon, \lambda) \in V \quad \text{and} \quad \varphi(\varepsilon, \lambda) = 0$$

is equivalent to

$$\varepsilon \in W \quad \text{and} \quad \lambda = k(\varepsilon, 0).$$

The mapping $\varepsilon \leftrightarrow k(\varepsilon, 0)$ is nothing else than what we called λ_d. Thus it results from (3.15) and (3.16) that

$$\partial \lambda_d(\varepsilon_0) \subset \frac{1}{\lambda_d(\varepsilon_0) \partial^2 f_d(\lambda_d(\varepsilon_0))}. \tag{3.17}$$

Hence, λ_d is a strictly increasing (local) lipeomorphism of $]0, \varepsilon_d^*[$ on $\mathbb{R}_+^*$ and it is easy to strengthen the inclusion (3.17) to equality. ■

Under the assumptions of theorem 3.3, ρ_d is locally Lipschitz and we deduce from (3.11) and (3.12) that

$$\begin{aligned} \partial \rho_d(\varepsilon) &= \frac{1 - [f_\varepsilon'(x_0; d) - f'(x_0; d)] \partial \lambda_d(\varepsilon)}{[\lambda_d(\varepsilon)]^2} \\ &= \frac{1}{[\lambda_d(\varepsilon)]^2} \left[1 - \frac{f_\varepsilon''(x_0; d, d)}{\partial^2 f_d(\lambda_d(\varepsilon))} \right] \quad \text{for all} \quad \varepsilon \in]0, \varepsilon_d^*[. \end{aligned} \tag{3.18}$$

Furthermore, the function θ_d studied in section 2 is $C^{1,1}$ on $]0, \varepsilon_d^*[$ and chain rules on generalized gradients [6], [8] make that

$$\partial^2 \theta_d(\varepsilon) = -\frac{1}{[\lambda_d(\varepsilon)]^3 \partial^2 f_d(\lambda_d(\varepsilon))} \quad \text{for all } \varepsilon \in]0, \varepsilon_d^*[. \tag{3.19}$$

Remark 1. If f_d is assumed to be C^2 on $\mathbb{R}_+^*$ with $(f_d)''(\lambda) > 0$ for all $\lambda > 0$, then λ_d is clearly C^1 on $]0, \varepsilon_d^*[$. In particular, we have that

$$\rho_d'(\varepsilon) = \frac{\langle \nabla^2 f(x_0 + \lambda_d(\varepsilon) d) d, d \rangle - f_\varepsilon''(x_0; d, d)}{[\lambda_d(\varepsilon)]^2 \langle \nabla^2 f(x_0 + \lambda_d(\varepsilon) d) d, d \rangle} \tag{3.20}$$

whenever f is C^2 in a neighbourhood of each point of $x_0 + \mathbb{R}_+^* d$. In such a case, it is symbolical to note that the sign of $\rho_d'(\varepsilon)$ is that of the difference between the 'exact second order directional derivative at the approximate point $x_0 + \lambda_d(\varepsilon) d$' and the 'approximate second order directional derivative at x_0'.'

Remark 2. Under the assumptions of theorem 3.3, it merely comes from (3.18) that a necessary and sufficient condition for ρ_d to be increasing on $(\underline{\varepsilon}, \bar{\varepsilon}) \subset]0, \varepsilon_d^*[$ is that

$$f_\varepsilon''(x_0; d, d) \leqslant \lambda^* \quad \text{for all } \lambda^* \in \partial^2 f_d(\lambda_d(\varepsilon)) \quad \text{and} \quad \varepsilon \in]\underline{\varepsilon}, \bar{\varepsilon}[.$$

A strictly inequality would secure that ρ_d is strictly increasing on $(\underline{\varepsilon}, \bar{\varepsilon})$.

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